REPORT
SUPER MARKET SALES PERFORMANCE DASHBOARD

1. Project Overview

I began this project by performing Exploratory Data Analysis (EDA) on the supermarket sales dataset. During the EDA phase, I examined data types, checked for missing values, identified duplicates, and explored distributions of key variables such as sales amount, product line, and rating. This helped in understanding patterns, cleaning the dataset, and selecting appropriate metrics for visualization.

2.Dataset Summary

- Total Rows: 1000
- Total Columns: 17
- Dataset Type: Supermarket sales transactions
- Numerical Columns: Unit price, Quantity, Tax, Total amount, Rating
- Categorical Columns: Branch, City, Customer type, Gender, Product line, Payment
- Date/Time Columns: Date, Time
- Missing Values: No missing values were present

3.Exploratory Data Analysis Using Python

Data Preparation and Cleaning
We began with data preparation and cleaning using Python.

- **Data Loading**: Imported the dataset using the pandas library.
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- **Initial Exploration**: Used df.info() to understand the structure of the dataset and df.describe() to generate summary statistics for numerical features.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice_ID               1000 non-null   object
 1   Branch                   1000 non-null   object
 2   City                     1000 non-null   object
 3   Customer type            1000 non-null   object
 4   Gender                   1000 non-null   object
 5   Product_line             1000 non-null   object
 6   Unit_price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax                      1000 non-null   float64
 9   Total                    1000 non-null   float64
 10  Date_                    1000 non-null   object
 11  Time_                    1000 non-null   object
 12  Payment                  1000 non-null   object
 13  cogs                     1000 non-null   float64
 14  gross_margin_percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
```

```
df.describe(include='all')
```

	Invoice_ID	Branch	City	Customer type	Gender	Product_line	Unit_price	Quantity	Tax	Total	Date_
count	1000	1000	1000	1000	1000	1000	1000.000000	1000.000000	1000.000000	1000.000000	1000
unique	1000	3	3	2	2	6	NaN	NaN	NaN	NaN	89
top	849-09-3807	A	Yangon	Member	Female	Fashion accessories	NaN	NaN	NaN	NaN	2019-07-02 00:00:00
freq	1	340	340	501	501	178	NaN	NaN	NaN	NaN	20
mean	NaN	NaN	NaN	NaN	NaN	NaN	55.672130	5.510000	15.379369	322.966749	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	26.494628	2.923431	11.708825	245.885335	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	10.080000	1.000000	0.508500	10.678500	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	32.875000	3.000000	5.924875	124.422375	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	55.230000	5.000000	12.088000	253.848000	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	77.935000	8.000000	22.445250	471.350250	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	99.960000	10.000000	49.650000	1042.650000	NaN

- **Missing Data Handling and Duplicate Values:**Checked for null values and duplicate values
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- **Column Standardization:**Renamed columns to ensure consistency, improve readability, and align with documentation standards

```
df.isnull().sum()
```

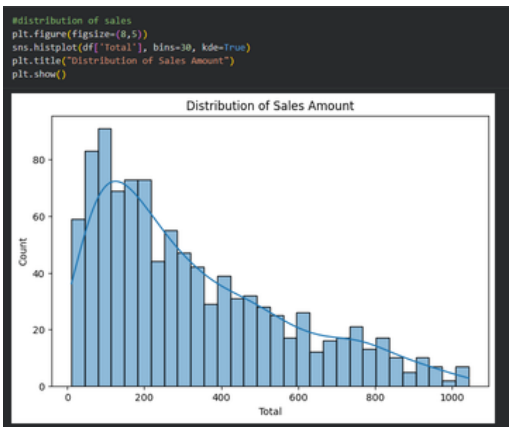
	0
Invoice_ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product_line	0
Unit_price	0
Quantity	0
Tax	0
Total	0
Date_	0
Time_	0
Payment	0
cogs	0
gross_margin_percentage	0
gross income	0
Rating	0

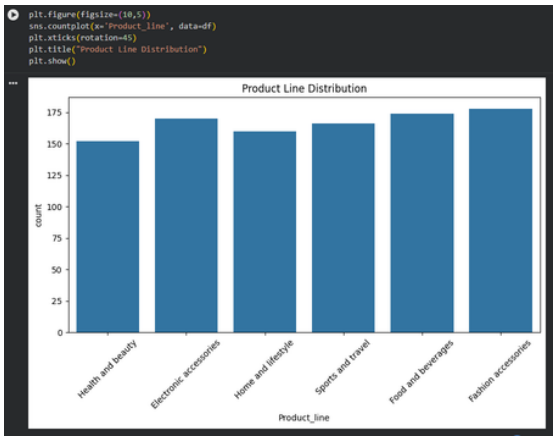
dtype: int64

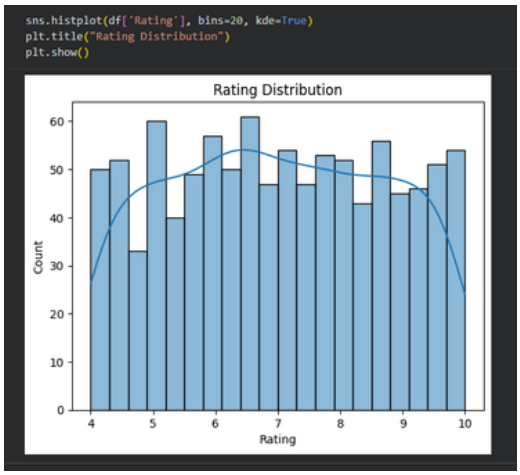
```
print("Total duplicates:", df.duplicated().sum())

duplicates = df[df.duplicated(keep=False)]
print(duplicates)

Total duplicates: 0
Empty DataFrame
Columns: [Invoice_ID, Branch, City, Customer_type, Gender, Product_line, Unit_price, Quantity, Tax_,
Index: []
```

```
df.columns=df.columns.str.replace(' ','_')
df.columns

Index(['Invoice_ID', 'Branch', 'City', 'Customer_type', 'Gender',
       'Product_line', 'Unit_price', 'Quantity', 'Tax_', 'Total', 'Date_',
       'Time_', 'Payment', 'cogs', 'gross_margin_percentage', 'gross_income',
       'Rating'],
      dtype='object')
```

Performed distribution analysis of sales, product lines, and customer ratings using Python







```python
print(df.describe())
print(df['Product_line'].value_counts())
print(df['City'].value_counts())
print(df['Branch'].value_counts())
print(df['Payment'].value_counts())
```

```
       Unit_price     Quantity         Tax_        Total        cogs  \
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.00000
mean     55.672130     5.510000    15.379369   322.966749   307.58738
std      26.494628     2.923431    11.708825   245.885335   234.17651
min      10.080000     1.000000     0.508500    10.678500    10.17000
25%      32.875000     3.000000     5.924875   124.422375   118.49750
50%      55.230000     5.000000    12.088000   253.848000   241.76000
75%      77.935000     8.000000    22.445250   471.350250   448.90500
max      99.960000    10.000000    49.650000  1042.650000   993.00000

       gross_margin_percentage  gross_income       Rating
count             1.000000e+03   1000.000000  1000.00000
mean              4.761905e+00     15.379369     6.97270
std               6.131498e-14     11.708825     1.71858
min               4.761905e+00      0.508500     4.00000
25%               4.761905e+00      5.924875     5.50000
50%               4.761905e+00     12.088000     7.00000
75%               4.761905e+00     22.445250     8.50000
max               4.761905e+00     49.650000    10.00000
Product_line
Fashion accessories       178
Food and beverages        174
Electronic accessories    170
Sports and travel         166
Home and lifestyle        160
Health and beauty         152
Name: count, dtype: int64
City
Yangon       340
Mandalay     332
Naypyitaw    328
Name: count, dtype: int64
Branch
A    340
B    332
C    328
Name: count, dtype: int64
Payment
Ewallet        345
Cash           344
Credit card    311
Name: count, dtype: int64
```

4.Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key Problem Questions

1.Total quantity of product sold inthe Supermarket

	sum bigint
1	5510

2.Total number of male and female customer

	gender character varying (10)	count bigint
1	Female	501
2	Male	499

3.Which Supermarket branch has generated the maximum revenue

	branch character varying (2)	total_revenue numeric
1	C	110568.86

4.The avg product rating of each branch

	city character varying (20)	branch character varying (2)	avg numeric
1	Yangon	A	7.0270588235294118
2	Mandalay	B	6.8180722891566265
3	Naypyitaw	C	7.0728658536585366

5.Total revenue of each branch

	branch character varying (2)	total_revenue numeric
1	A	106200.57
2	B	106198.00
3	C	110568.86

6.Date on which the highest sales were recorded

```
print(df.describe())
print(df['Product_line'].value_counts())
print(df['City'].value_counts())
print(df['Branch'].value_counts())
print(df['Payment'].value_counts())
```

7.Most preferred payment method in each city

	city character varying (20)	payment character varying (30)	payment_count bigint
1	Mandalay	Ewallet	113
2	Naypyitaw	Cash	124
3	Yangon	Ewallet	126

8.Best Performing Branch on the basis of total revenue and rating

	branch character varying (2)	total_revenue numeric	avg_rating numeric
1	C	110568.86	7.0728658536585366